

In real life, we often need to measure relationships in order to predict future trends, quantify driving factors and identify key variables. All these can be transformed into build the right model based on training dataset, even though the training dataset may be greatly different from testing dataset.

The basic idea of DP is to "stop parameter selection at a 'scale' of noise level. When residual is less than noise, the solution begins to fit the noise, and at this point, the process stops."

The basic idea of LP is to "minimize the total error by achieving a balance between approximation error and stability error."